UVEITIS

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Uveitis = inflammation of the uveal tract. High yield in theory + viva!

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Classification + Anatomy of Uvea

Uvea = iris + ciliary body + choroid. Know the SUN classification!

DEFINITION

Inflammation of the uveal tract (iris, ciliary body, choroid) — may extend to retina, vitreous, sclera.

Anatomical Classification (SUN Working Group)

Type	Primary Site	Key Features
Anterior	Iris / ciliary body	Most common (60-70%), iritis/iridocyclitis
Intermediate	Vitreous / pars plana	Snowbanking, snowball opacities
Posterior	Choroid / retina	Choroiditis, retinitis, chorioretinitis
Panuveitis	All layers involved	Severe — VKH, Behcet, sympathetic ophthalmia

Clinical Classification

Duration	Onset	Course
Acute	Sudden onset	< 3 months, symptomatic
Chronic	Insidious onset	> 3 months, may be asymptomatic
Recurrent	Repeated episodes	Inactive periods in between

Common Etiologies (by Type)

Type	Common Causes
Anterior	HLA-B27, JIA, Herpes, TB, Sarcoidosis, Fuchs heterochromic
Intermediate	Multiple sclerosis, Sarcoidosis, Pars planitis (idiopathic)
Posterior	Toxoplasmosis (#1 infective), TB, CMV, Toxocara
Panuveitis	VKH, Behcet disease, Sympathetic ophthalmia, Sarcoidosis

This comparison table
is THE most asked SAQ!
Must learn by heart!

Granulomatous vs Non-Granulomatous Uveitis

Feature	Granulomatous	Non-Granulomatous
Onset	Insidious / chronic	Acute / sudden
Pain	Mild / absent	Marked pain + photophobia
KPs	Mutton-fat (large, greasy)	Fine, dusty, small
Iris nodules	Koeppe + Busacca nodules	Usually absent
Flare	Moderate	Marked
Synechiae	Broad-based, posterior	Fine, posterior
Course	Chronic, relapsing	Acute, self-limiting
Causes	TB, Sarcoid, Syphilis, VKH	HLA-B27, JIA, Idiopathic

★ **KPs (Keratic Precipitates): Inflammatory cells deposited on corneal endothelium.**
Settle in Arlt's triangle (inferior cornea) due to convection currents in AC.
Stellate KPs (evenly spread) = Fuchs heterochromic iridocyclitis

Iris Nodules in Granulomatous Uveitis

Nodule Type	Location & Features
Koeppe nodules	At pupillary margin of iris (most common)
Busacca nodules	On iris stroma surface (more specific for granulomatous)
Berlin nodules	At angle of anterior chamber

Important Disease Associations

Disease	Uveitis Type
Ankylosing spondylitis	Acute anterior, non-granulomatous, HLA-B27+
Sarcoidosis	Chronic, granulomatous, bilateral, any type
Behcet disease	Panuveitis, hypopyon, recurrent, HLA-B51
JIA (juvenile)	Chronic anterior, band keratopathy, girls
Toxoplasmosis	Posterior — focal necrotizing retinochoroiditis

Clinical Features + Complications

Complications of uveitis = exam gold! Know the dreaded complications!

Signs of Anterior Uveitis

Sign	Detail
Circumcorneal congestion	Ciliary injection (deep vessels, violaceous)
KPs on endothelium	Fine (non-gran) or mutton-fat (gran)
Cells + flare in AC	Cells = WBCs, Flare = protein exudation
Posterior synechiae	Iris adheres to lens -> irregular pupil
Festooned pupil	Irregular pupil due to broken synechiae
Hypopyon	Pus in AC (sterile) — Behcet, HLA-B27
Miosis	Small pupil due to sphincter spasm
Low IOP	Due to decreased aqueous production

Complications of Uveitis

ANTERIOR

- Complicated cataract
- Secondary glaucoma
- Band-shaped keratopathy
- Posterior synechiae -> seclusio pupillae
- Iris bombe -> angle closure

POSTERIOR

- Cystoid macular edema (CME)
- Epiretinal membrane
- Retinal detachment
- Choroidal neovascularization
- Optic disc edema

Seclusio vs Occlusio Pupillae

Seclusio pupillae: 360-degree posterior synechiae (ring synechiae) -> aqueous trapped

Occlusio pupillae: pupil covered by inflammatory membrane -> iris bombe -> secondary glaucoma

Investigations + Management

Treatment = steroids + cycloplegics! Simple but must know details!

Investigations (Tailored Approach)

Investigation	Purpose
CBC, ESR, CRP	Baseline inflammatory markers
Mantoux / QuantiFERON	Tuberculosis screening
Chest X-ray	TB, Sarcoidosis (hilar lymphadenopathy)
VDRL / FTA-ABS	Syphilis screening
HLA-B27	Ankylosing spondylitis, reactive arthritis
Serum ACE + lysozyme	Sarcoidosis
Toxoplasma IgG/IgM	Toxoplasmosis (posterior uveitis)
FFA / OCT	Macular edema, choroiditis evaluation

Management of Uveitis

Drug Class	Details
Topical steroids	Prednisolone acetate 1% — mainstay of anterior uveitis
Cycloplegics	Atropine 1% / Homatropine — prevents synechiae, relieves pain
Systemic steroids	Oral prednisolone — for posterior/pan uveitis
Immunosuppressants	Methotrexate, Azathioprine, Cyclosporine — steroid-sparing
Biologics	Anti-TNF (Adalimumab) — refractory cases, Behcet
Specific Rx	Anti-TB, antiviral, anti-toxoplasma as per etiology

- Rule out infective causes BEFORE starting steroids!
- Cycloplegics: prevent posterior synechiae + relieve ciliary spasm (pain relief)
- Steroid tapering: gradually reduce — sudden stop causes rebound inflammation
- Monitor IOP: steroids can cause secondary open angle glaucoma

★ **Sympathetic Ophthalmia**

Bilateral granulomatous panuveitis after penetrating trauma to one eye.

Exciting eye (injured) + Sympathizing eye (fellow). Rx: enucleation within 2 weeks.

Exam Strategy + High Yield

Uveitis = guaranteed
LAQ + SAQ every year!
Comparison tables = gold!

Most Repeated LAQs (10–12 marks)

1. Classify uveitis. Describe anterior uveitis — features & management
2. Differentiate granulomatous from non-granulomatous uveitis (table!)
3. Describe complications of anterior uveitis and their management
4. Write about sympathetic ophthalmia — etiology, features, treatment

Most Repeated SAQs (5–6 marks)

1. Keratic precipitates — types and significance
2. Posterior synechiae — causes and complications
3. Hypopyon — causes (sterile vs infective)
4. Band-shaped keratopathy — cause and treatment
5. Vogt-Koyanagi-Harada syndrome
6. Behcet disease — ocular features

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common type of uveitis	Anterior uveitis (iridocyclitis)
Mutton-fat KPs	Granulomatous uveitis (TB, sarcoid)
Stellate KPs (evenly spread)	Fuchs heterochromic iridocyclitis
Hypopyon uveitis	Behcet disease (sterile)
HLA-B27 uveitis	Acute anterior, non-granulomatous
Band keratopathy in children	JIA-associated chronic uveitis
Candle-wax dripping (taches)	Sarcoidosis (retinal periphlebitis)
Headlight in fog	Toxoplasma posterior uveitis
#1 infective cause posterior uveitis	Toxoplasmosis
Dalen-Fuchs nodules	Sympathetic ophthalmia / VKH
Exciting eye vs sympathizing eye	Sympathetic ophthalmia

♥ **Uveitis treatment gives people their vision back, Sammy!**
Every inflamed eye you treat = a life changed. Keep learning!